

DECODED EXPLAINED

31st July, 2026

ISRO's NavIC System Can No Longer Provide Navigation Data Independently: Govt.

ISRO's NavIC system can no longer provide navigation data independently: Govt

Anonna Dutt
New Delhi, July 30

FOR THE first time since positioning data from one of its satellites was lost in March, the government has accepted in the parliament that the seven-satellite navigation constellation (NavIC) can no longer independently aid people in navigation.

The Indian Space Research Organisation (ISRO) earlier this year informed that IRNSS-1F satellite, which was launched in March 2016, completed its designed mission life of 10 years and the on-board atomic clock stopped functioning.

"...presently the NavIC constellation has three satellites for providing PNT (Positioning, Navigation, Timing) services — IRNSS-1B, IRNSS-1I and NVS-01. To provide basic positioning service, minimum four operational satellites are required in orbit. Hence, the NavIC constellation cannot provide standalone positioning service, however timing service is functional," union minister of state for space Dr Jitendra Singh replied in a written answer in Parliament.

The reply said that the armed forces continue to use NavIC signals as part of the multi-constellation GNSS — Global Navigation Satellite System that uses data from US' GPS, EU's Galileo,

and Russia's Glonass among others to navigate.

When asked whether it compromises accuracy for navigation or real-time mapping, the reply stated: "Since NavIC is being used in multi constellation environment for most of the use-cases, there is no vulnerability for the users."

While end-user applications were developed for utilising NavIC signals in aviation, shipping, railways, and even in everyday cell phones, all of these devices carry a GNSS receiver, meaning they can continue to get positioning data from other constellations.

The reply also stated: "Next NavIC satellite NVS-03 is ready for launch and two more satellites NVS-04 and NVS-05 are in advanced stage of realisation."

The constellation can still provide accurate time signals for digital devices as well as broadcast emergency messages.

The Indian Regional Navigation Satellite System (IRNSS) — also referred to with the operational name of Navigation with Indian Constellation (NavIC) — was planned to be a seven-satellite system to provide positioning data over the Indian subcontinent and 1500 km around it. Ever since its inception, however, it has been plagued by problems affecting its positioning data.

ISRO's NavIC System Can No Longer Provide Navigation Data Independently: Govt.

The Indian Regional Navigation Satellite System (IRNSS), operationally known as Navigation with Indian Constellation (NavIC), is India's indigenous satellite navigation system.

Extend

It was designed to provide accurate positioning information to users in India and up to 1500 km beyond its borders, which constitutes the Primary Service Area.

In addition, an Extended Service Area covers the region between the primary area and the rectangle bounded by 30° South to 50° North latitude and 30° East to 130° East longitude.

Need for NavIC:

The project was initiated to ensure strategic autonomy in navigation services.

During the 1999 Kargil conflict, the US denied India access to GPS data, highlighting the risks of relying on foreign systems. To address this vulnerability, India approved the NavIC project in 2006.

ISRO's NavIC System Can No Longer Provide Navigation Data Independently: Govt.

First Generation (IRNSS Series): The first generation of NavIC satellites includes the IRNSS-1 series, launched between 2013 and 2018.

Key satellites include IRNSS-1A, 1B, 1C, 1D, 1E, 1F, 1G, and the replacement satellite IRNSS-1I.

IRNSS-1H (2017), intended to replace IRNSS-1A, failed to reach orbit due to heat shield separation failure.

Three satellites of the constellation are placed in geostationary orbit, and four satellites are placed in inclined geosynchronous orbit

The ground network consists of control centre, precise timing facility, range and integrity monitoring stations, two-way ranging stations, etc.

Second Generation (NVS Series): The NVS series represents the second generation of NavIC satellites, developed to improve reliability and expand capabilities.

It includes NVS-01, NVS-02, and the planned (NVS-03, NVS-04, and NVS-05).

ISRO's NavIC System Can No Longer Provide Navigation Data Independently: Govt.

Other Countries Having Satellite Navigation Systems

United States: Global Positioning System (GPS)

Russia: GLONASS (GLObalnaya NAVigatsionnaya Sputnikovaya Sistema)

European Union: Galileo

China: BeiDou

Regional Navigation Systems

India: NavIC (IRNSS)

Japan: Quasi-Zenith Satellite System (QZSS), which augments GPS signals over Japan.

Key Differences in Satellite Constellations

GPS, GLONASS, and Galileo: Operate with over 20 satellites in Medium Earth Orbit (~20,000 km).

BeiDou: Uses over 40 satellites in a combination of Medium Earth Orbit and Geosynchronous Orbit (~35,000 km).

NavIC and QZSS: Have fewer satellites and mainly operate in higher geosynchronous orbits, focusing on regional coverage rather than global navigation.

To meet the navigation requirements of civil aviation, ISRO and Airports Authority of India (AAI) have implemented the GPS Aided Geo Augmented Navigation – GAGAN as a satellite based augmentation system (SBAS) for the Indian airspace.

ISRO's NavIC System Can No Longer Provide Navigation Data Independently: Govt.

With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements:

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

This question was previously asked in
Official UPSC Civil Services Exam
2018 Prelims Part A

[Download PDF](#)

[Attempt Online](#)

[View all UPSC Civil Services Papers >](#)

1. 1 only
2. 1 and 2 only
3. 2 and 3 only
4. None

Which one of the following countries has its own Satellite Navigation System?

This question was previously asked in
UPSC Civil Services Prelims
2023: General Studies (SET - A...

[Download PDF](#)

[Attempt Online](#)

[View all UPSC Civil Services Papers >](#)

1. Australia
2. Canada
3. Israel
4. Japan

Temporary Respite: June's Strong Industrial Performance Is Not a Sign of Things to Come

Temporary respite

June's strong industrial performance is not a sign of things to come

The June 2026 data for the **Index of Industrial Production** has come as a pleasant surprise. The 23-month high growth rate of 7.3% of the overall index was unexpected given the significant economic headwinds from the West Asia crisis and a deficient monsoon. Some of the growth was due to a low base, since the performance last June was the worst in nearly a year, but the numbers nevertheless indicate resilience. The manufacturing sector accelerated due to a dual push from the domestic and the external sectors. Consumer durables growth remained above 7% for the second consecutive month, and the non-durables sector saw growth quicken to a six-month high. At the same time, data from the Commerce Ministry showed that merchandise exports grew 15.5% in June, revealing demand from abroad. The capital goods sector also saw double-digit growth. The sector has grown in double digits in eight out of the last 10 months. This bodes well for the economy, but also reveals some risks. Capital creation, primarily by the government, has been the single consistent engine of growth in the post-pandemic years. It needs to keep firing. There are too many uncertainties and headwinds for other engines such as exports and domestic consumption to consistently take up the load. Other areas of growth are also far too seasonal. The electricity sector grew at a 25-month high in June, largely in response to a heat wave in several parts of the country. The mining sector, too, snapped a four-month streak of contraction and grew in June. Yet, this will likely be temporary since the progressing monsoon will disrupt mining activities.

Economists have also warned that the monsoon's deficiency will inevitably hit rural demand in the months ahead, which will see the consumer-facing sectors slumping again. Hopes of a ceasefire in West Asia are also evaporating. This is leading to considerable volatility in oil prices, which is sending ripples of uncertainty through the economy. This will play out over the coming months. Planned investments will remain pending, purchases will be deferred, and savings will increasingly overshadow consumption. The economy has displayed resilience so far, but this resilience will be strained as its duration increases. Further, people's needs are such that simple resilience will not suffice for long. If the external environment is going to remain unfavourable, the government will need to think radically about how to jumpstart the domestic economy. In the meantime, it will have to keep pumping money into capital expenditure even as other fiscal pressures mount.

Temporary Respite: June's Strong Industrial Performance Is Not a Sign of Things to Come

Index of Industrial Production

Published monthly by the National Statistical Office (NSO) under the Ministry of Statistics and Programme Implementation (MoSPI), it serves as a reliable barometer for the overall health of the industrial sector.

The IIP base year has been officially updated from 2011-12 to 2022-23, making it the latest major macroeconomic indicator to be revised in 2026 alongside the GDP.

Temporary Respite: June's Strong Industrial Performance Is Not a Sign of Things to Come

About Core Sectors:

Core industries serve as the structural backbone for all other industrial activities. The combined weight of these core industries in the broader Index of Industrial Production (IIP) is now 32.88% (under the new 2022-23 base), updated from the previous 40.27% under the 2011-12 series.

Addition of Iron Ore: The revised ICI has expanded the list of core industries from eight to nine by including Iron Ore, recognizing its extensive use in steel manufacturing and its vital contribution to industrial development.

Consequently, the revised ICI now covers Coal, Crude Oil, Natural Gas, Refinery Products, Fertilizers, Steel, Cement, Electricity, and Iron Ore.

Electricity has the highest weight (30.93%), followed by Refinery Products (22.57%), Steel (17.58%), Crude Oil (7.43%), Coal (5.59%), Iron Ore (4.90%), Cement (4.41%), Natural Gas (3.84%), and Fertilizers (2.73%).

Temporary Respite: June's Strong Industrial Performance Is Not a Sign of Things to Come

In India, in the overall Index of Industrial Production, the Indices of Eight Core Industries have a combined weight of 37.90%. Which of the following are among those Eight Core Industries?

1. Cement
2. Fertilizers
3. Natural gas
4. Refinery products
5. Textiles

Select the correct answer using the codes given below:

This question was previously asked in
Official UPSC Civil Services Exam
2012 Prelims Part A

[Download PDF](#)

[Attempt Online](#)

[View all UPSC Civil Services Papers >](#)

1. 1 and 5 only
2. 2, 3 and 4 only
3. 1, 2, 3 and 4 only
4. 1, 2, 3, 4 and 5

ISRO & Japanese Scientists Review Mission Chandrayaan-5 Preparation

ISRO & Japanese scientists review mission Chandrayaan-5 preparation

Anjali Marar
Bengaluru, July 30

A JAPANESE space delegation on Wednesday held discussions with their counterparts at Indian Space Research Organisation (ISRO) to take stock of the progress being made on Chandrayaan-5 — a joint mission to the Moon scheduled for 2028.

Scientists from Japan's Aerospace Exploration Agency (JAXA) visited the UR Rao Satellite Centre in Bengaluru on Wednesday. The two countries have been working on Chandrayaan-5 also known as Lunar Polar Exploration Mission (LUPEX), a first-of-its-kind joint rover mission to the Moon. For this project, ISRO is building the lander module, while JAXA is developing the rover, the one which will come out of the lander module after landing and move around on the surface of the Moon, surveying the topography and collecting data.

The mission will take off on a Japanese rocket, H3, and is planned to carry seven scientific instruments for carrying out different kinds of experiments. The

Gaganyaan-1 in progress

Bengaluru: The assembly, integration and testing of the Crew and Service Modules of the Gaganyaan-1 human spaceflight mission are nearing completion, Science and Technology Minister Jitendra Singh said in a written reply in Lok Sabha on Wednesday. He detailed the mission's status, and completed tasks in his reply to Jaipur MP Rao Rajendra Singh.

Gaganyaan-1 uncrewed mission was originally scheduled for launch in late 2025. It was later pushed to the first quarter of 2026. Singh had, during the Winter Session of House, 2025, said that the G-1 launch would take place before March 2026. But, ISRO is yet to release the revised launch date. **ENS**

National Aeronautics and Space Administration (NASA) of the US and the European Space Agency (ESA) are also participating in this mission, and will be mounting their payloads on the rover. The ESA is developing the

mass spectrometer and NASA is developing the neutron spectrometers. Chandrayaan-5 is planned to stay on the Moon for 100 days i.e. 3 and five months.

During their Wednesday meeting, scientists of ISRO and JAXA discussed the technical co-ordination required for Inter Spacecraft Communication of Chandrayaan-5. This mission aims to study water and water-ice both on the lunar surface and the subsurface.

The two agencies took stock of the onboard payloads and on the possible launch window, which is likely to be in 2028. Last week, JAXA successfully performed the second Payload Operations Simulation at Tsukuba Space Center. The complex operational scenario, involving both spacecraft bus and payload operations, were tested in parallel.

Chandrayaan-5 was approved by the Government in March 2025 in the backdrop of India's landing on the lunar south pole with the Chandrayaan-3 mission in August 2023. ISRO's immediate next mission to the moon will be the Chandrayaan-4 mission.

ISRO & Japanese Scientists Review Mission Chandrayaan-5 Preparation

India and Japan have jointly planned the upcoming Chandrayaan-5 (LUPEX) mission to study the Moon's surface and the subsurface for locating lunar water and ice deposits

Chandrayaan-5 also known as LUPEX (Lunar Polar Exploration) is a joint lunar mission by ISRO (India) and JAXA (Japan's space agency), scheduled for 2027-28 launch on Japan's H3 rocket.

Rover will be developed by JAXA and lander by ISRO.

Thank You